

Kavach MAC_CODE issue

1. Algorithm Applied

The following algorithm was implemented:

1. KAVACH_ID is read from the .env file as a fixed unit identifier (e.g. 50002).
2. Loco_ID is read from bytes 7–9 of the packet.
3. The Key Index is computed as $(\text{KAVACH_ID} + \text{Loco_ID}) \% 2$.
4. A Key Index of 0 selects Authentication Key 1, and a Key Index of 1 selects Authentication Key 2.
5. The packet's Date field (bytes 13–15, DD MM YY) is used to look up the correct key set in KMS_KEY_SETS.
6. The key set row is chosen where $\text{VALID_FROM} \leq \text{packet date} \leq \text{VALID_TO}$.
7. KEY_1 and KEY_2 are decrypted using AES-256-GCM with AAD of the form $\text{KMS_KEY_SETS}|\text{KEY_n}|\{\text{key_set_id}\}$.
8. The selected Authentication Key (KA) is used directly as the AES key — no session-key derivation and no random numbers are involved.
9. The MAC input consists of the bytes from Message Type through the last payload byte.
10. SOF, MAC_CODE, and CRC are excluded from the MAC input.
11. The MAC input is zero-padded to a multiple of 16 bytes.
12. AES-128 CBC mode is used.
13. The IV is 16 zero bytes.
14. The final 16-byte cipher block is taken as the CBC-MAC output.
15. The first 4 bytes of the final cipher block are used as the MAC_CODE.
16. The computed MAC_CODE is compared with the 4 bytes stored at $\text{packet}[-8:-4]$ to verify the packet.

2. Result

The implemented calculation produced the following result.

```
(base) khushboo@khushboo:~/Documents/CRIS-official/new-kms-integ/KMS/MAC_Generation$ python check_mac.py
=====
CBC-MAC computation
=====
Key           : 7AD4637D0CB980DC70BC7AA5D43C2B41
MAC input     : 180023054701000F61780118091A0F05080300106A00106800106C
Input length  : 27 bytes
Padded length : 32 bytes (5 zero bytes added)
IV            : 16 zero bytes
Algorithm     : AES-128 CBC-MAC, take first 4 bytes
-----
Full 16-byte MAC : B5C927D634D7D9FA8367A91EB00C38EA
Computed MAC_CODE: B5C927D6
OEM           MAC_CODE: BB01BD09
-----
❌ MISMATCH – computed MAC_CODE does not equal OEM MAC_CODE.
   (OEM MAC does not appear anywhere in the 16-byte block)
=====
```

```
(base) khushboo@khushboo:~/Documents/CRIS-official/new-kms-integ/KMS/MAC_Generation$ python kavach_mac.py
=====
KAVACH MAC tool – generate / verify
Key selection: (Server_ID + Loco_ID) % 2
=====
1) Generate MAC_CODE
2) Verify   MAC_CODE
Choose [1/2] > 2
Packet hex > bbbb180025020b01000f61780116091a0e0c0b0300fb00f000210754003103539eea87fcf947fd
-----
Packet type      : 0x18 (Onboard KAVACH Health)
Server ID (.env) : 50002
Loco ID (packet) : 65551
Packet date      : 22-Sep-2026
Packet length    : 39 bytes
MAC input length : 29 bytes
Key selection : (50002 + 65551) % 2 = 1 -> Authentication Key 2
-----
Received MAC : 539EEA87
Computed MAC : 9EB97E0F
CRC in packet: FCF947FD
-----
❌ MAC is INVALID
=====
```